DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #01G: PATHS & SUBPATHS (PART 7 — GRANULAR AUDIT OF GRAND DAO WAVE 2A: `SHADOW_SAMURAI`, `NINPO_SHADOW`, `NINPO_ELEMENTAL`, `ARENA_GLORY`, `FORTUNE_GAMBLER`, `TIDE_VOYAGER`, `SPIRIT_HUSBANDRY`, `ANCESTRAL_RITUAL` & ALL 32 SUBPATHS)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Series Position (`01C–01H` Granular 43-Path / 286-Subpath Audit)**:
  - *(Filtered against `JS2_V2_4_0_FIX_TRACKER_TH.md` so previously fixed `v2.4.0` items such as `DA-001` stunned global finisher and `DA-002`/`DA-005` wager stone minting are NOT repeated; every issue below is verified live in `JS2-v2.6.0`.)*
  - **`01G` (This Document)**: Group 5 — **8 Martial, Shinobi, Arena, Fortune, Maritime, Agricultural & Ancestral Grand Dao Paths (`SHADOW_SAMURAI` [4], `NINPO_SHADOW` [4], `NINPO_ELEMENTAL` [4], `ARENA_GLORY` [4], `FORTUNE_GAMBLER` [4], `TIDE_VOYAGER` [4], `SPIRIT_HUSBANDRY` [4], `ANCESTRAL_RITUAL` [4] — 32 Subpaths total)**.

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order & Closed-Loop Economy**: Gambler wagers must use personal `npc.inv.stones` (never unauthorized `W.sect.stones` treasury raids!), Ancestral offerings must consume real stock items, and Spirit Husbandry harvests must respect plot capacity and `W.world.resourcePool`.
3. **100% Emergent NPC Autonomy**: Samurai train at the sword dojo, Shinobi infiltrate and reposition, Gladiators seek public tournaments, Mariners fish and sail, Farmers cultivate

through harvests, and Shamans venerate real Ancestor Tablets recorded in `W.sect.ancestors`.

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1. [PATH-SAMURAI-01] Path #28: SHADOW_SAMURAI (วิถีซามูไรเพลงดาบไร้เงา — 4 Subpaths)

- **Subpaths**: `IAI_FLASH_SLICER`, `MURAMASA_BLOOD_EDGE`, `UNBROKEN_IRON_STANCE`, `KENJUTSU_SWORD_SAINT`.
- **Core Rule**: Zero Dantian / Zero Qi (`CULT.qiMax(n) === 0`), strict `SWORD` & `BLADE` Bushido discipline, powered by `IAI_FOCUS` (สมาธิคมดาบ) and physical Stamina (`h.state.energy`).
- **Six Live Zero-Qi Pipeline Leaks in `JS2-v2.6.0`**:
  1. **Zero-Resource Ultimate Skill Spam (`SK.qiCost` in `js/skills.js` & `js/combat.js`)**:
     - Active skills compute `qiCost = Math.round(CULT.qiMax(a) * costPct / 100)`. Because `CULT.qiMax(a) === 0`, `qiCost` evaluates to **`0`**!
     - `COMBAT.useSkill` deducts `0` Qi, `0` `energy`, and `0` `IAI_FOCUS`—allowing `SHADOW_SAMURAI` to spam Tier-7 `LEG_V230_SHADOW_SAMURAI` and `TECH_UNI_V230_*` ultimates completely for free!
     - **Fix**: When `CULT.isZeroQiSamurai(a)` casts an active skill with `costPct > 0`, require and deduct **`staminaCost = Math.max(2, Math.round(costPct * 0.35))` from `a.h.state.energy`** (or `15..25` `IAI_FOCUS` for signature Iai strikes)!
  2. **Buying Useless Qi-Restore Pills (`js/cpeco.js` & `js/external_market.js`)**:
     - Block `SHADOW_SAMURAI` NPCs from spending CP or personal stones on `qiRestore`-only pills; redirect their preference to Stamina/Body-Recovery pills and Whetstones (`ITEM_REPAIR_STONE`)!
  3. **Mortal Initiation Contradiction (`MORTAL_INITIATION` in `js/cultivation.js`)**:
     - Mortals studying `SHADOW_SAMURAI` currently meditate at `shrine`/`meditation_hall` gathering "Heaven-and-Earth Qi into their Dantian" scaled by Spirit Roots! Route `SHADOW_SAMURAI` (and `ARENA_GLORY`) initiation to `training_ground` (ฝึกตวัดดาบพันครั้งขัดเกลาสมาธิ), scaling progress by `strength`, `reflex`, `focus`, and `willpower` rather than Spirit Roots!
  4. **Mantra `+matk` Waste & Zero-Qi Lockouts in `QINGGONG` / `DEFENSIVE_ARTS` (`stats.js:261, 470`, `cultivation.js`)**:

    - Route `SHADOW_SAMURAI` Mantra bonuses into `+atk / +def / +crit` (cross-ref `03C`), and pay `SHADOW_BLINK` / `DEFENSIVE_ARTS` upkeep from `energy` instead of `qi` (cross-ref `03D`, `03E`).
5. **Ranged Laser Parry Bug on `UNBROKEN_IRON_STANCE` (`combat.js:244`)**:
    - `parryCounter` (`48% ATK` counter-slash) currently triggers even when hit by an archer or mage **25 tiles away**! Restrict `parryCounter` counter-slash to melee range (`unitDist(a, d) <= 2.5`), while deflecting `-30%` projectile damage from ranged hits!
6. **Missing Subpath Tradeoffs (`IAI_FLASH_SLICER` & `MURAMASA_BLOOD_EDGE`)**:
    - `IAI_FLASH_SLICER`: Missing or having an Iai finisher parried/dodged should apply a 4-tick `exposed` (ช่องว่างขณะเก็บดาบเข้าฝัก) window, honoring its `supremeRisk`!
    - `MURAMASA_BLOOD_EDGE`: Add cursed-blade thirst (`-1.2%` HP/round in combat until drawing blood, granting `+14%` lifesteal against `bleed` targets)!

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2. `[PATH-SHINOBI-02]` Paths #29 & #30: `NINPO_SHADOW` (4 Subpaths) & `NINPO_ELEMENTAL` (4 Subpaths)

2.1 `NINPO_SHADOW` (วิถีนินจาเร้นกายและมนต์มายา — 4 Subpaths)

- **Subpaths**: `CLONE_MIRAGE_WEAVER`, `SMOKE_MIST_ASSASSIN`, `CONCEALED_WEAPON_STRIKER`, `SHADOW_BIND_STRATEGIST`.
- **Verified Combat & World Gaps**:
    1. **`shadowClone` (`1 HP` Clone) AoE Interaction & Missing Clone-Break Vulnerability**:
        - Lore states: *"สร้างร่างแยกเงามายา (1 HP) มารับการโจมตีแทน · หากร่างแยกเงาถูกทำลายหมดและถูกโจมตีวงกว้าง จะได้รับความเสียหายเพิ่มขึ้น"*.
        - Ensure `p.aoe === true` sweeps destroy **all** active `1 HP` shadow clones simultaneously (passing `65%` splash damage through to the real body) and apply a 6-tick `exposed` debuff when the final clone shatters—giving AoE sweepers (`SPEAR` / `FAN` / `GRAVITY_DAO`) a crisp tactical counter against clone ninjas!
    2. **`CONCEALED_WEAPON_STRIKER` (สายนินจาดาวกระจายและอาวุธลับ)**:
        - Grant `+1` bonus `armedConsumables` slot for throwing knives/poison darts and `+20%` `KW_STATUS_EXTEND` on `NEEDLE`/`BLADE` attacks.
    3. **Missing Sect Espionage & Night-Ops Role (`sect.js` / `expedition.js`)**:
        - `JianghuSect2_v2.3.0_Bug_Audit_Report.md` (Sec 3) noted `SHINOBI_OPS` was never implemented. Wire `NINPO_SHADOW` directly

into existing `intel` / `scout` posts and dungeon/siege prep (`+28%` enemy sect scouting accuracy, `-30%` spy capture risk, `+20%` night combat evasion)!

2.2 NINPO_ELEMENTAL (วิถีนินจาอักขระคาถาและยันต์ระเบิด — 4 Subpaths)

- **Subpaths**: `EXPLOSIVE_TAG_TRAPPER`, `FIVE_SEAL_ELEMENTALIST`, `SUBSTITUTION_ESCAPE`, `CHAKRA_SCROLL_SEALER`.
- **Three Concrete Fixes**:
  1. **`SUBSTITUTION_ESCAPE` (คาถาสลับร่างท่อนไม้) Fails to Reposition Out of Melee (`js/combat.js`)**:
     - When `substitutionReady` triggers cheat-death, the ninja swaps with a wooden log **but stays at the exact same `(x, y)` coordinate surrounded by melee enemies**! Call `projectSafePosition` (`+2.2` tiles away from attacker + clear `closedOnBy = null`) upon substitution!
  2. **`FIVE_SEAL_ELEMENTALIST` Mudra Elemental Primer & Interrupt Penalty**:
     - Deposit `+0.8U` Elemental Aura on targets per mudra cast (feeding `COMBAT.ELEMENTAL_FUSION`!), and if hit by `stun`/`silence` while casting, apply `-20%` `matk` for 6 ticks (มุทราขาดตอน).
  3. **`EXPLOSIVE_TAG_TRAPPER`**: Plant delayed Paper-Bomb Echoes (`DELAYED_ECHO` `FIRE` AoE after 3 ticks) when enemies close into melee!

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3. [PATH-ARENA-03] Path #43: ARENA_GLORY (วิถีเกียรติยุทธ์ลานประลอง — 4 Subpaths)

- **Subpaths**: `GRAND_TOURNEY_CHAMPION`, `DUELIST_HONOR_BLADE`, `CROWD_ROAR_BRAWLER`, `BANNER_GLORY_MARSHAL`.
- **Simulation & Context Contradictions**:
  1. **Gladiators Sit at the Shrine Instead of Sparring & Entering Tournaments!**:
     - Boost `ARENA_GLORY` utility in `ai.js` / `cognition.js` for `A.practice`, `pair` (`spar`), `duel`, `SECT.assembly` tournaments, and `SECT.WORLD_MARTIAL_CONFERENCE` (`+35` preference!), awarding real `CULT.gain` and `RENOWN_FLAME` on every public bout won!
  2. **Audience-Blind `CROWD_ROAR_BRAWLER` & Zero Cowardice Penalty**:
     - Lore states: **"ยิ่งต่อสู้ท่ามกลางสายตาผู้คน เพลิงเกียรติยศยิ่งลุกโชน · ไม่ถนัดการลอบโจมตีหรือการหนีจากสมรภูมิ"**.
     - Compute `hasAudience = (f.kind === 'tourney' || f.kind === 'leadership' || f.kind === 'raid' || (f.A.length + f.B.length >= 4))`; boost `CROWD_ROAR_BRAWLER` & `GRAND_TOURNEY_CHAMPION` `RENOWN_FLAME` gain by `+40%` when `hasAudience` is true (and `-15%` in isolated 1v1 wilderness ambushes).

- **Honor Code**: Raise flee reluctance (`fleeThreshold * 0.55`) for `ARENA_GLORY`; if forced to flee (`n.fled = true`), collapse `RENOWN_FLAME` to `0` and add `+15` humiliation `stress` (อัปยศหนีทัพต่อหน้ายุทธภพ)!
- **`DUELIST_HONOR_BLADE`**: `+16%` damage & `+12` parry accuracy in honorable `1v1` engagements (`enemies.length === 1` or vs `RIVAL`/`NEMESIS`).

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4. `[PATH-GAMBLER-04]` Path #41: `FORTUNE_GAMBLER` (วิถีโชคลาภและการเดิมพันชะตา — 4 Subpaths)

- **Subpaths**: `JADE_DICE_HIGH_ROLLER`, `FATE_COIN_FLIPPER`, `KARMA_ROULETTE_DEALER`, `ALL_IN_DESPERADO_BLADE`.
- **Emergent Ownership & Subpath Gaps (Post-`v2.4.0` Fix)**:
  1. **Gambling With Public Sect Funds (`W.sect.stones`) Without Treasury Authority!**:
     - Currently, autonomous `FORTUNE_GAMBLER` wagers pull directly from **`W.sect.stones`** even if the gambler is a Rank-1 Outer Disciple!
     - **Fix**: Autonomous personal wagers MUST stake **`n.inv.stones` (capped at `25%` of `n.inv.stones`)** against either a consenting high-`sensationSeeking` peer (`peer.inv.stones`, pure NPC-to-NPC transfer!) or traveling tavern/merchant tables (`W.world.resourcePool.stones` $\leftrightarrow$ `n.inv.stones`, strictly bounded by pool balance!). Only `HOUSE_BANKER` / `KARMA_ROULETTE_DEALER` appointed as `merchant`/`treasury` may run an authorized Sect Escrowed House Pool!
  2. **Distinct Subpath Wager Mechanics in Combat**:
     - `FATE_COIN_FLIPPER`: Each round flips Destiny Coin (`W.rng.next()`)—Heads grants `+24% critDmg & +12 cast`; Tails grants `+15% eva` but `-10% def` (ช่องว่างเมื่อดวงตก).
     - `KARMA_ROULETTE_DEALER`: Randomly swaps 1 positive buff from an enemy onto an ally and 1 debuff from an ally onto an enemy (`ICD: 16 ticks`, boss-safe)!
     - `ALL_IN_DESPERADO_BLADE`: At `hp <= 30%`, consumes all `STAKE` (`>= 40`) for one guaranteed-crit desperation cleave, followed by `-20 energy` fatigue!

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5. [PATH-TIDE-FARM-ANCESTOR-05] Paths #39, #42 & #38: TIDE_VOYAGER (4), SPIRIT_HUSBANDRY (4) & ANCESTRAL_RITUAL (4)

5.1 TIDE_VOYAGER (วิถีนาวิกสมุทรและเหมันต์วารี — 4 Subpaths)

- **Subpaths**: ABYSSAL_TIDE_SPEAR, FROST_MAELSTROM_CALLER, DRAGON_SHIP_NAVIGATOR, DEEP_PEARL_APOTHECARY.
- **Disconnected From `js/fishing.js` & Terrain/Weather**:
  - Wire TIDE_VOYAGER into `js/fishing.js`: +35% rare spirit-fish & underwater pearl-chest discovery (DEEP_PEARL_APOTHECARY), +25% damage vs river/sea beasts (ABYSSAL_TIDE_SPEAR), and real `CULT.gain` + TIDE_FLOW gauge from fishing and river/coastal expeditions!
  - Wire DRAGON_SHIP_NAVIGATOR into `EXPED.createParty` (`waterRouteMult * 0.72`, +30% carry capacity on water routes).
  - Enforce `supremeRisk`: +20% skill Qi cost when fighting in `volcano` / `desert` regions or `crimson_sun` drought unless `rain` or a WATER_ICE domain is active; +18% TIDE_FLOW generation near water tiles or during `rain`/`storm`!

5.2 SPIRIT_HUSBANDRY (วิถีเกษตรวิญญาณและพฤกษาโอสถ — 4 Subpaths)

- **Subpaths**: IMMORTAL_HERB_PLANTER, IRON_THORN_VINE_WEAVER, SPIRIT_GRAIN_SOVEREIGN, PARASITIC_SPORE_CULTIVATOR.
- **Disconnected From `A.farm` & `wild_resources.js`**:
  - Route SPIRIT_HUSBANDRY disciples to **cultivate through real agricultural labor (`A.farm` & `gatherWildResource`)**: completing a farm/herb cycle grants **`3.5× CULT.gain`**, charges GROWTH_CYCLE, and increases `grain` (SPIRIT_GRAIN_SOVEREIGN +30% harvest) or rare spirit-herb yield (IMMORTAL_HERB_PLANTER +22%, bounded by `resourcePool`)!
  - IRON_THORN_VINE_WEAVER: Protects herb plots from `wildlife.js` pest raids and roots melee attackers (8 ticks, 16-tick ICD, boss-resistant).
  - PARASITIC_SPORE_CULTIVATOR: Applies `sporeSeed` in combat that drains 4% enemy Qi per tick to heal nearby allies!
  - Enforce `supremeRisk`: +15% incoming `burn` / FIRE damage when unshielded!

5.3 ANCESTRAL_RITUAL (วิถีพิธีบรรพชนและไขกระดูกมรดก — 4 Subpaths)

- **Subpaths**: MARROW_HERITAGE_SHAMAN, ANCESTRAL_TABLET_CHANTER, HEROIC_SPIRIT_SPEARMAN, CHRONICLE_TOTEM_GUARDIAN.
- **Disconnected From Real Sect Ancestors (`W.sect.ancestors`) & Multi-Generational Lineage (`Gen 2–3`)**:

- Currently `ANCESTRAL_RITUAL` grants static flat stats even on Day 1 when `W.sect.ancestors` is completely empty!
- **Authentic Living-History Scaling**:
  - Scale `RITE_FERVOR` cap and `ANCESTRAL_RITUAL` aura dynamically with **real recorded Ancestor Tablets (`W.sect.ancestors.length` + ascended immortals + `W.sect.sectChronicle` milestones)**, capped cleanly (`+3%` per revered tablet up to `+24%`)!
  - Add an autonomous `AI.ACTS.venerateAncestors` rite at `shrine` (consuming 1 `meal` or `incense`/herb offering from `W.sect.stock` every 15 days to keep `W.sect.ancestralVenerationUntil` active; if neglected > 30 days, `RITE_FERVOR` decays by `-30%`, honoring "หากป้ายวิญญาณบรรพชนขาดการสักการะ พลังหนุนนำจะอ่อนกำลังลง")!
  - `MARROW_HERITAGE_SHAMAN`: Grants `+25%` study speed on Sect Heritage Arts (`isSectHeritage`) and `+20%` bloodline awakening potency for Gen 2–3 offspring!

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6. Implementation & Verification Checklist for GPT / Gemini (`01G`)

- ☐ **`SHADOW_SAMURAI` Zero-Qi Integrity**: Deduct `energy` (stamina) / `IAI_FOCUS` on active skill casts when `qiMax === 0`; block Qi-pill purchases in `cpeco.js`/`external_market.js`; route mortal initiation to `training_ground` physical/focus stats; restrict `parryCounter` slash to melee range; add missed-Iai `exposed` window and `MURAMASA` blood-thirst.
- ☐ **`NINPO_SHADOW` & `NINPO_ELEMENTAL`**: Make AoE sweeps pop `1 HP shadowClone` stacks + apply `exposed` on last clone break; reposition `(x, y)` on `SUBSTITUTION_ESCAPE`; add `FIVE_SEAL` elemental primer + interrupt penalty; wire night/espionage bonuses to `scout`/`intel`.
- ☐ **`ARENA_GLORY` & `FORTUNE_GAMBLER`**: Prefer sparring/tournaments in AI, scale `RENOWN_FLAME` with audience presence, collapse `RENOWN_FLAME` on fleeing; restrict autonomous gambler stakes to personal `n.inv.stones` (`Debit === Credit`) + wire distinct subpath coin/roulette/desperado mechanics.
- ☐ **`TIDE_VOYAGER`, `SPIRIT_HUSBANDRY` & `ANCESTRAL_RITUAL`**: Connect `TIDE_VOYAGER` to `fishing.js` + water/desert modifiers; connect `SPIRIT_HUSBANDRY` to `A.farm` `CULT.gain` + `sporeSeed`/fire weakness; connect `ANCESTRAL_RITUAL` to real `W.sect.ancestors` tablets, offering upkeep, and `isSectHeritage` / Gen 2–3 lineage bonuses.